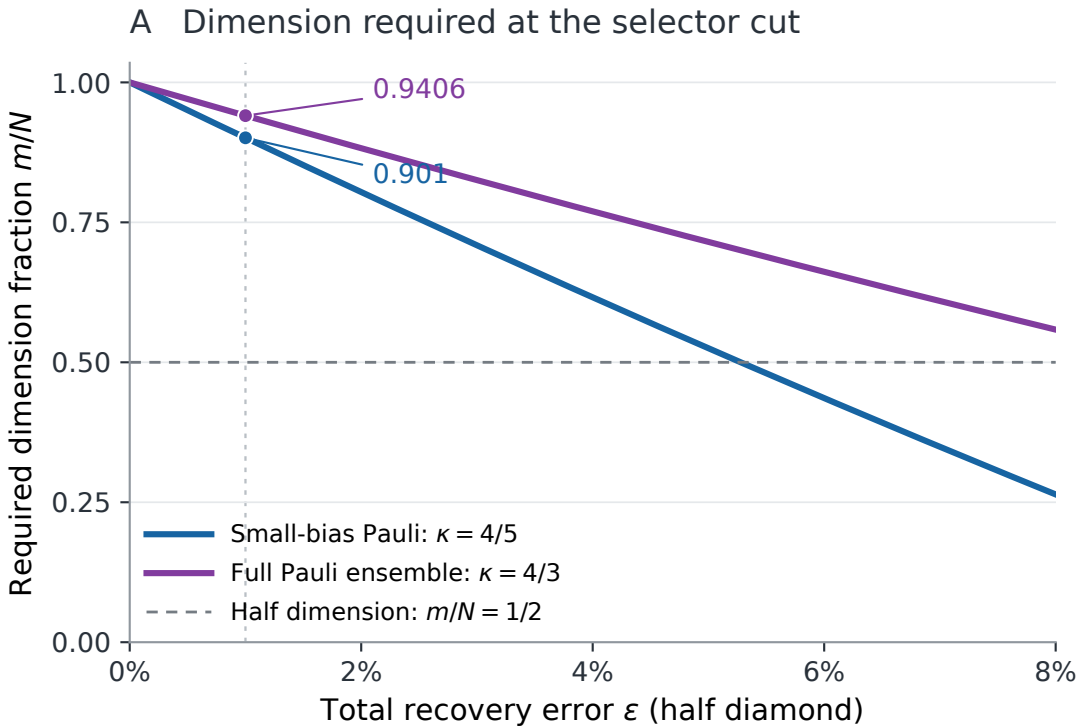
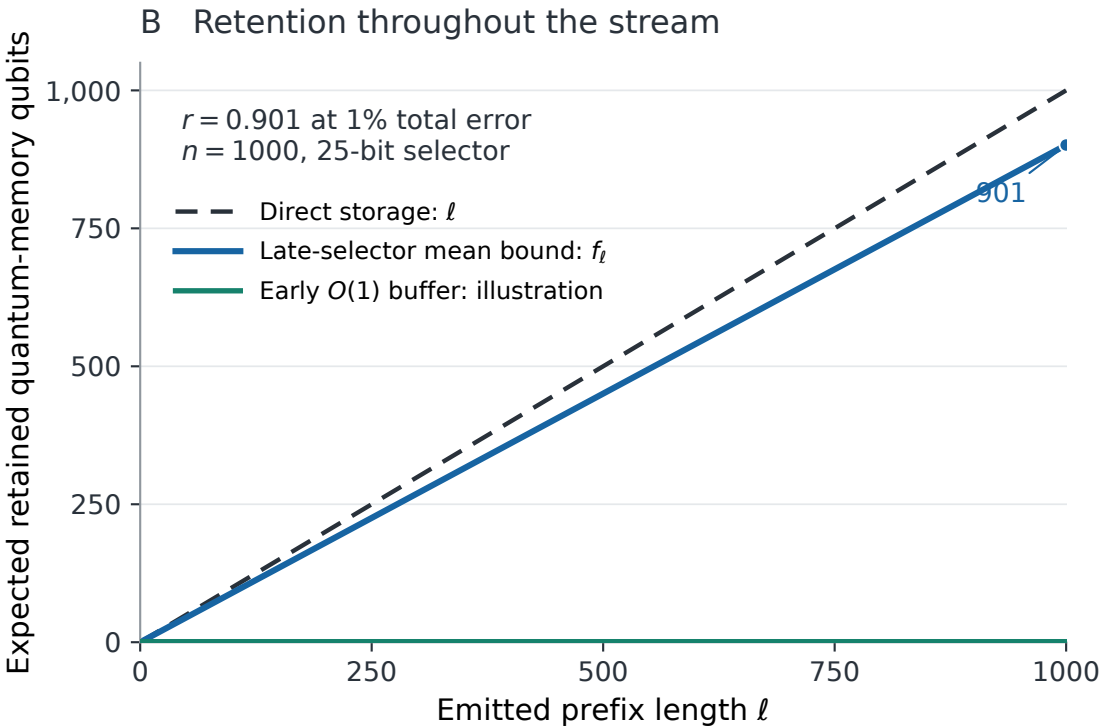
Late metadata forces quantum retention

Analytic theorem bounds | No device simulation





Cumulative bound at 1,000 gauge arrivals: 450,950.23 qubit-slots; direct storage uses 500,500.

Mean bounds use the Bell-input marginal and allow variable classical branches. Early-buffer line shows scaling only; its height is not an apparatus count.

One slot is an arrival interval τ. Delay t_{hdr} before any header information adds at least $901.901\,t_{\mathrm{hdr}}$ qubit-time. Protection overhead is additional.